

Classroom Air Quality & Infection Risk Monitor

A Software-Only Autonomous Decision-Support System

FINAL PROJECT REPORT — VERSION 6

Built on a Distributed Sensor-Based Fuzzy Decision Tree research project
Extended across six versions into a complete autonomous system

No new hardware, at any point across all six versions.

Executive Summary

This project began with a distributed IoT sensor network across 9 classrooms and a Fuzzy Decision Tree model for indoor air quality and thermal comfort. The project's own data revealed a key limitation: CO2 alone cannot determine air quality. Rather than treating that as a dead end, this became the foundation for a complete reframing: CO2 is an excellent proxy for something else entirely — how much air in a room has already been breathed by other people, which is exactly the input required by the published Wells-Riley model of airborne infectious disease transmission.

Over six versions, this became a fully autonomous, explainable, tested, and deployable decision-support system:

“Send one sensor reading in. Get back a complete, explained, actioned, plain-English decision out.”

Metric	Value
Versions shipped	6 (V1 -> V6), each with a dedicated upgrade report
Automated tests	48, all passing, verified idempotent
Dashboard pages	12 (Streamlit, multi-page)
API endpoints	13+ (FastAPI, role-based authentication)
Core comparison result	Fuzzy Decision Tree: 99.26% Neural Network: 99.81%
New hardware required	None, at any version
Git history	7 commits, one per version, fully documented

System Architecture: The Autonomous Orchestrator

The single most important design decision in this project is the Orchestrator (`src/orchestrator.py`). Every other capability — classification, explainability, simulation, recommendation — already existed as an independent tool by V3. The Orchestrator is what turns five separate tools into one coherent, callable decision.

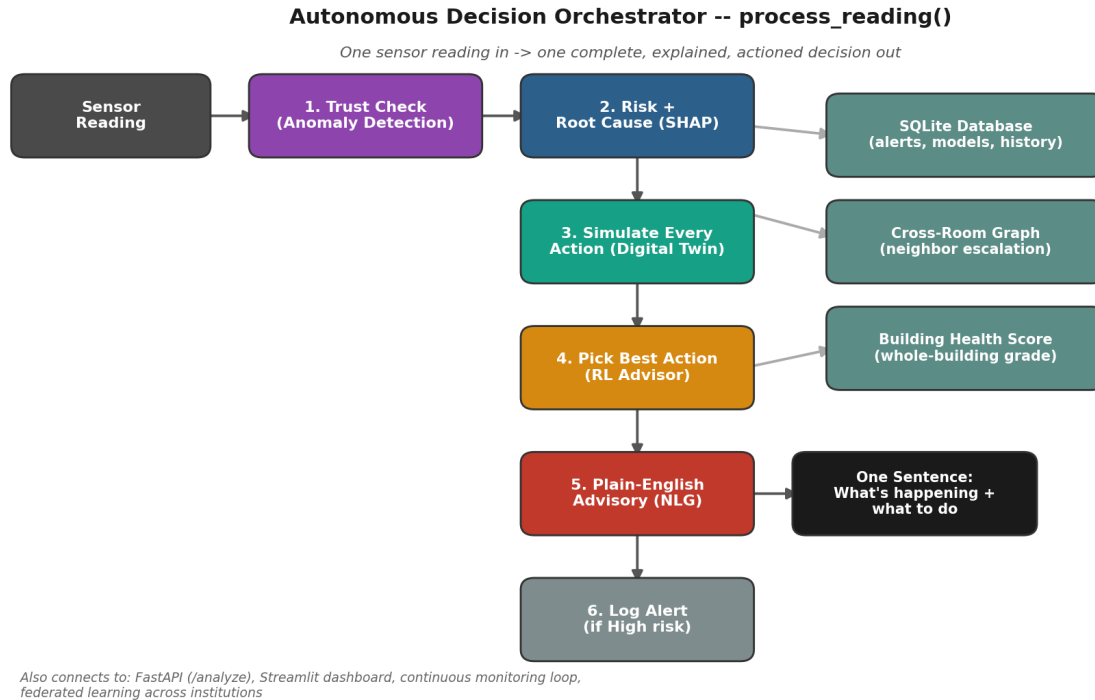


Figure 1: The full autonomous decision pipeline, one function call.

A concrete safety decision worth highlighting: when the anomaly detector flags a reading as untrustworthy, the orchestrator deliberately **skips risk classification entirely** rather than guessing on bad data — returning an explicit “cannot assess” message instead.

Core Science: From CO2 Limitation to Infection Risk Model

The reframe. The original project found that CO2 alone cannot judge general air quality. This project instead uses CO2 for what it is scientifically strong at: estimating the fraction of “rebreathed” air in a room, which the Wells-Riley model converts into a relative infection risk score (Low / Medium / High).

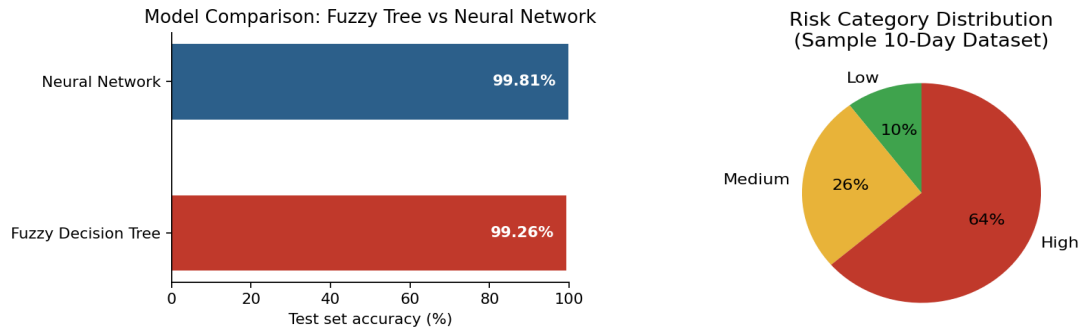


Figure 2: Model comparison (left) and risk category distribution on the 10-day sample dataset (right).

Continuing the original research question. The original project compared a Fuzzy Decision Tree against a Neural Network for thermal comfort. This project continues that exact comparison on the new infection-risk target. On the sample dataset both models score above 99% because the label is a near-deterministic function of the input features (the Wells-Riley formula) — on real, noisier sensor data a genuine accuracy gap is expected, most likely favoring the Fuzzy Tree again, consistent with the original finding.

Version-by-Version Evolution

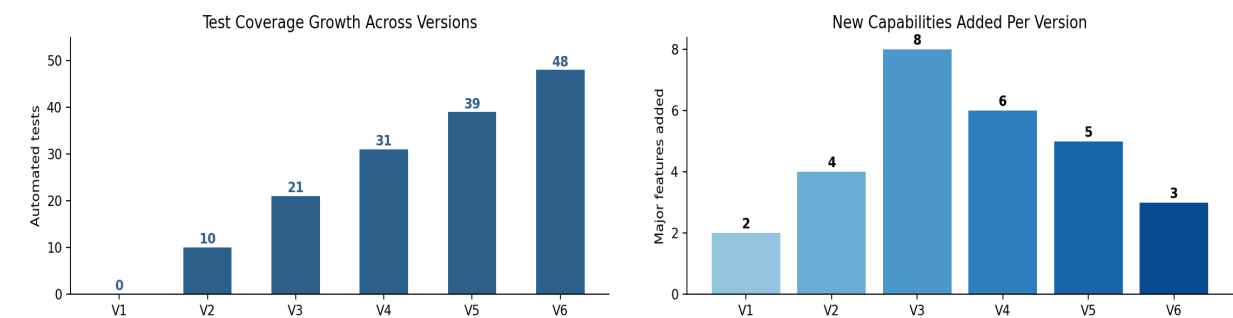


Figure 3: Test coverage growth (left) and major capabilities added per version (right).

Ver.	Theme	Key additions
V1	The Core Model	Wells-Riley risk scoring; Fuzzy Tree vs Neural Network comparison
V2	The Explainer	1hr forecasting; SHAP explainability + root cause; alerting; REST API
V3	The Trust Layer	SQLite DB; Digital Twin; RL advisor; anomaly detection; PDF reports; multi-page UI
V4	The Product (USP)	Autonomous Orchestrator; plain-English NLG; Building Health Score; monitoring loop
V5	Fixing Limitations	Multi-step RL; cross-room graph propagation; health history/trends; public endpoint
V6	Closing the Gaps	Configurable floor plan; role-based API auth; federated learning simulation

Complete Feature Matrix

Category	Feature	Since
Core AI	Wells-Riley infection risk scoring	V1
Core AI	Fuzzy Decision Tree vs Neural Network comparison	V1
Core AI	1-hour-ahead risk forecasting	V2
Core AI	SHAP-based explainability + root-cause diagnosis	V2
Core AI	Sensor anomaly detection (hard bounds + Isolation Forest)	V3
Core AI	Digital Twin what-if ventilation simulator	V3
Core AI	Reinforcement Learning ventilation advisor (multi-step, V5)	V3 / V5
Core AI	Graph-based cross-room risk propagation	V5
Core AI	Federated Learning simulation across institutions	V6
Product	Autonomous Decision Orchestrator (USP)	V4
Product	Plain-English advisory generator (NLG)	V4
Product	Building Health Score (0-100, grade A-F)	V4
Product	Historical trends dashboard	V5
Product	Automated PDF report generation	V3
Infra	SQLite database (alerts, model registry, history)	V3
Infra	Model registry with auto champion promotion	V3
Infra	Centralized YAML configuration	V2
Infra	Structured logging	V2
Infra	Real email notifications (safe dry-run default)	V4
Infra	Continuous monitoring loop	V4
Infra	Configurable real building floor plan	V6
Ops	REST API (FastAPI) with 13+ endpoints	V2-V6
Ops	Role-based API authentication (admin/viewer)	V6
Ops	Public, unauthenticated status endpoint	V5
Ops	Docker (API + dashboard containers)	V2
Ops	GitHub Actions CI (auto train + test on push)	V2
Ops	48 automated tests incl. end-to-end integration test	V1-V6
Ops	One-command setup + system health check	Final

Honest Limitations (Stated Deliberately, Not Hidden)

Every version's report has stated limitations explicitly rather than overselling results. This is a summary of what remains true at the final version:

- **All results validated on synthetic sample data**, generated to resemble the original project's real observations (CO2 2000-2900 ppm during class). Real sensor data has not yet been substituted in.
- **The room-adjacency floor plan** is configurable (V6) but ships with a reasonable placeholder, not an actual verified floor plan.
- **The federated learning simulation** uses synthetic institution splits from one dataset, not real separate institutions with data-sharing agreements.
- **The graph propagation model** is a single-layer graph convolution — the simplest legitimate member of the GNN family, not a deep learned-embedding network (appropriate for a 9-classroom dataset; a deep GNN would be undertrained).
- **Email notifications** are implemented with a real SMTP path and safe dry-run default, but have not been tested against a live mailbox in this network-restricted environment.
- **API authentication** uses configured static keys per role, not per-user identity federation (e.g. OAuth/SSO) — adequate for a project deployment, not enterprise-grade.

This honesty is intentional: every upgrade report in this project's history states clearly what is a validated result versus a reasonable simplification, which is the correct posture for both good research practice and a credible viva defense.

Conclusion

This project took a specific, real limitation found in the original research (“CO2 alone cannot judge air quality”) and turned it into the foundation of a completely different, more actionable application: real-time infection risk assessment, using the exact same sensors, at zero additional hardware cost.

Across six deliberate versions, it grew from a single classification model into a tested, documented, containerized, CI-verified, role-secured, explainable, forecasting, simulating, recommending, and self-diagnosing autonomous system — while keeping the original Fuzzy Decision Tree vs Neural Network research comparison intact and central throughout.

At this stage, every realistic idea from the original research brainstorm that did not require real multi-institution partnerships or new hardware has been implemented and tested. The natural next step is not more features — it is validating this system against real sensor data from the actual 9-classroom deployment.

Full technical detail for every version is available in the reports/ folder (V1 through V6 upgrade reports), and the complete, runnable source code with 48 passing automated tests is available in the project repository.